

Python Placement Roadmap

Python · Pandas · NumPy · SQL · Scikit-learn

Weekly Study Schedule

Day	Focus
Monday	🐍 Python + 🗄️ SQL
Tuesday	🐍 Python + 📊 Pandas + 🧮 NumPy
Wednesday	🐍 Python + 🗄️ SQL + 🤖 Scikit-learn
Thursday	🐍 Python + 📊 Pandas
Friday	🐍 Python + 🗄️ SQL + 🧮 NumPy
Saturday	🐍 Python + 🤖 Scikit-learn
Sunday	🕒 Mixed Interview Challenge — Python + 2 questions from SQL/Pandas/NumPy/Scikit-learn

🐍 Python

🔴 High Priority — Must Know

- Lists, tuples, sets, dictionaries
- List / dict / set comprehensions
- Functions
 - *args, **kwargs
 - Default arguments
 - Scope
- Lambda
- map(), filter(), sorted()
- String manipulation
- Exception handling
 - try / except / finally / raise
- File handling
 - Read / write, with open()
- OOP
 - Class / object, constructor
 - Inheritance, polymorphism, encapsulation
- Iterators & generators
- Decorators
- Shallow vs. deep copy
- Mutable vs. immutable
- is vs ==
- References and memory behavior

🟡 Medium Priority

- zip(), enumerate()
- any() / all()
- collections — Counter, defaultdict, deque
- datetime
- Regular expressions

- Modules & packages, `__name__ == "__main__"`
- JSON handling
- Basic type hints

Lower Priority

- `reduce()`
- Context managers — deeper implementation
- Advanced decorators
- Advanced dunder methods
- Metaclasses

Pandas — Priority for AI/ML/Data/DE

● High Priority — Must Know

- Series & DataFrame
- Creating DataFrames
- Reading/writing — `read_csv()`, `read_excel()`, `to_csv()`
- `head()`, `tail()`, `shape`, `columns`, `dtypes`
- `info()`, `describe()`
- Selecting columns
- `loc[]` and `iloc[]`
- Filtering rows — multiple conditions, `isin()`
- `isna()`, `notna()`
- `fillna()`, `dropna()`
- `drop()`, `rename()`, `astype()`
- Sorting
- `value_counts()`
- `unique()`, `nunique()`
- `groupby()`
- Aggregations — `sum`, `mean`, `count`, `min`, `max`, `agg`
- `merge()`, `join()`, `concat()`
- `apply()`
- String operations
- Date/time operations
- Duplicates
- Index handling

● Medium Priority

- `transform()`
- `map()`, `replace()`
- `melt()`
- `pivot()` / `pivot_table()`
- MultiIndex
- Rolling/window operations
- Categorical data
- Memory optimization
- Chunking large CSVs

● Lower Priority

- Advanced MultiIndex
- Advanced categorical operations
- Advanced performance optimization
- Custom extension features

NumPy

● High Priority

- `ndarray`
- Creating arrays — `zeros()`, `ones()`, `arange()`, `linspace()`
- Shape & dimensions, `reshape()`
- Indexing, slicing, boolean masking
- Data types
- Vectorization & broadcasting
- Aggregations — `sum`, `mean`, `min`, `max`, `std`
- Axis concept

- Element-wise operations
- Matrix operations — transpose
- Random numbers
- Copy vs. view

🌀 Medium Priority

- `concatenate()`, `stack()`, `split()`
- Sorting/searching
- Linear algebra — dot, matrix multiplication, inverse, determinant
- `where()`, handling NaN
- Performance/vectorization concepts

🌀 Lower Priority

- Advanced broadcasting
- Advanced linear algebra
- Structured arrays
- Advanced memory/layout concepts

SQL — For ML + GenAI + Data Science + Data Engineering

● High Priority — Must Know

- SELECT / WHERE, DISTINCT
- ORDER BY, LIMIT
- GROUP BY, HAVING
- Aggregate functions
- CASE
- NULL handling, COALESCE
- String & date functions
- INNER / LEFT / RIGHT JOIN
- Self join, multiple joins
- Subqueries, correlated subqueries
- CTEs
- Window functions — ROW_NUMBER(), RANK(), DENSE_RANK(), LAG(), LEAD(), PARTITION BY
- UNION / UNION ALL, EXISTS
- Constraints — primary/foreign keys, indexes
- Transactions, ACID
- Views, stored procedures, triggers
- Normalization
- Query optimization basics

● Medium Priority

- Recursive CTE
- Advanced window functions
- Execution plans
- Composite indexes
- Transactions & isolation levels in depth
- Deadlocks
- Temporary tables
- Stored procedures/functions in depth

● Lower Priority

- Advanced database internals
- Partitioning/sharding in depth
- Advanced query optimizer internals

Scikit-learn

● High Priority

- Train/test split
- Feature/target separation
- Preprocessing — StandardScaler, MinMaxScaler, OneHotEncoder, LabelEncoder, ColumnTransformer
- Pipelines
- Linear Regression, Logistic Regression
- Decision Tree, Random Forest
- KNN, SVM basics, Naive Bayes
- K-Means, hierarchical clustering basics
- Model evaluation
 - Classification: accuracy, precision, recall, F1, ROC-AUC
 - Regression: MAE, MSE, RMSE, R²
- Cross-validation, GridSearchCV, RandomizedSearchCV
- Overfitting/underfitting, bias/variance
- Feature selection basics

● Medium Priority

- PCA
- Ensemble methods — Gradient Boosting, Random Forest tuning
- XGBoost/LightGBM concepts
- Imbalanced datasets — SMOTE, class weights
- Learning curves
- Feature importance
- Calibration

● Lower Priority

- Advanced clustering
- Advanced dimensionality reduction
- Advanced ensemble techniques
- Advanced model selection